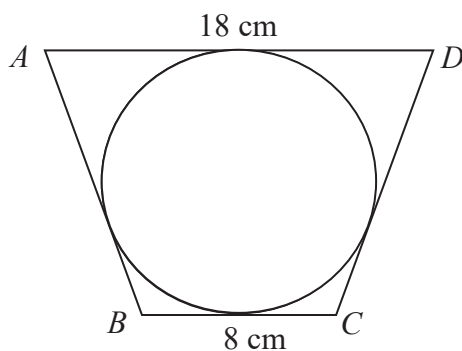


- (b) Given $ABCD$ is a trapezium with $AD = 18$ cm, $BC = 8$ cm and $AB = DC$. A circle is inscribed in the trapezium as shown.



- (i) Show that the length of AB is 13 cm, stating your reason(s) clearly.

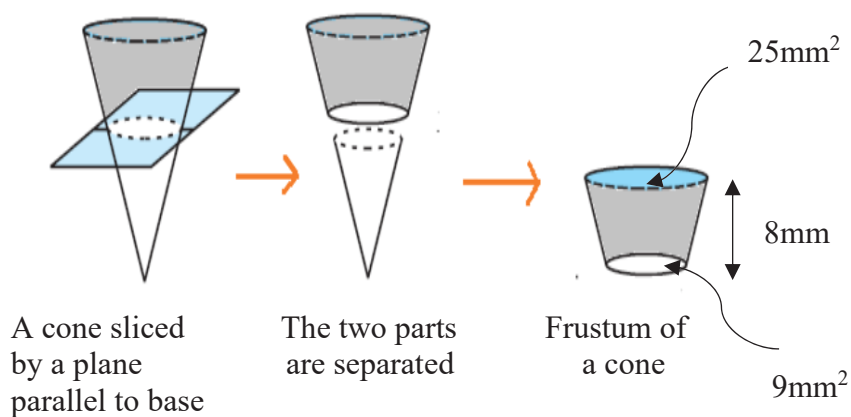
Answer

[2]

- (ii) Calculate the radius of the circle.

Answercm [2]

- 8 (a) A solid cone is cut into a smaller cone and a frustum as shown in the diagram.



The height of the frustum is 8mm. The area of the two circular bases are 9 mm² and 25 mm² respectively.

- (i) Show that the height of the smaller cone is 12 mm.

Answer

- (ii) Find the volume of the frustum.

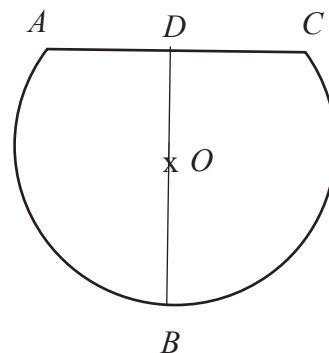
[2]

Answermm³ [3]

- (iii) Find the curved surface area of the smaller cone.

Answermm² [3]

- (b) In the figure, $ADCB$ is a major segment of a circle, centre O and radius 7 cm. $BD = 12$ cm. AC is perpendicular to the line BOD . Find the perimeter of the major segment.



Answercm [4]